PACE OCI Remote-Sensing Reflectance for Phytoplankton Community Classification and Short-Horizon Bloom-Risk Estimation

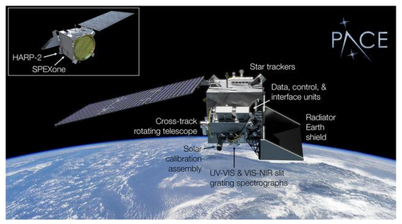
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Motivation

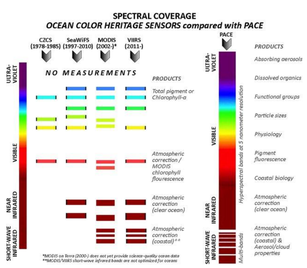
What is PACE:

NASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) mission launched on February 8, 2024, carrying the Ocean Color Instrument (**OCI**): a hyperspectral sensor measuring light from ultraviolet to shortwave infrared.

Since launch, PACE data has already been used to distinguish different phytoplankton communities from space and to support harmful algal bloom detection and fishery management.



The PACE observatory.
Visible features of OCI (and spacecraft) identified in the main panel.



The spectral coverage of ocean color heritage sensors compared to PACE.

- Image Credit: Werdell, P. et al., "The Plankton, Aerosol, Cloud, Ocean Ecosystem Mission: Status, Science, Advances," Bulletin of the American Meteorological Society 100(9):1775-1794, April 2019.

Background

Why Ocean Color Matters

- **Rrs(λ)**, Remote-sensing reflectance, measures sunlight reflected from the ocean. Since phytoplankton groups absorb and reflect light differently, their spectral signatures can reveal which communities are present, not just biomass. Distinguishing these groups is important for monitoring **harmful algal blooms (HABs)**.
- Heritage sensors like MODIS and SeaWiFS measure Rrs at only about 10 wavelengths, enough to estimate overall biomass, but not enough to resolve fine spectral differences between phytoplankton groups. PACE's OCI measures Rrs continuously across hundreds of wavelengths, offering far more spectral detail to work with.

Objective: Test whether PACE's dense satellite reflectance data contains a learnable signal for phytoplankton community type and short-term bloom risk, and **whether that signal can forecast near-future conditions using only past satellite data**.

Significance: Most satellite-based ocean monitoring can detect *that* biomass is present, but not *what* ***kind* of phytoplankton or *whether risk is rising***. A method built on satellite data alone could **support earlier, broader-scale bloom-risk monitoring.**

Proposed Method

Data & Matching

- In-situ phytoplankton labels (from SeaBASS, PVST, PACE-PAX) were matched by date and location to satellite reflectance data, pairing *what the satellite saw* with *what was actually in the water*
- **Currently, only a small number of in-situ samples are available with matched PACE coverage**, not enough to robustly test the PACE-based approach directly

Four Prediction Tasks

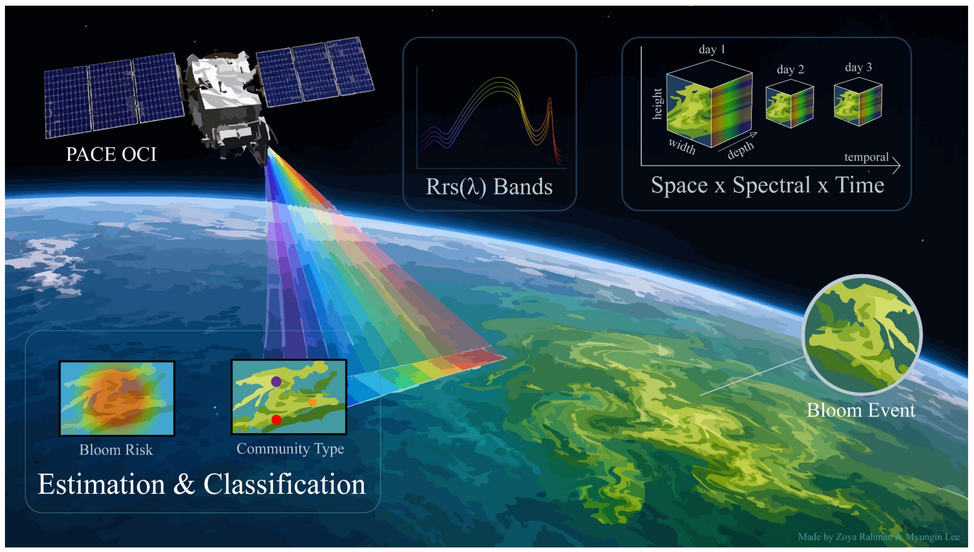
- **Current community type | Current bloom risk | Future community type | Future bloom risk**

Leakage-Safe Forecasting

- Future-task models only use satellite data from *before* the date being predicted, ruling out the model "peeking" at data from after the event

Model Comparison

- Several compact neural network architectures were tested, from simple single-image models (**small 2D**) to models using short time sequences (**small 3D**) and 3D spatial-temporal structure (**factorized**)
- All models were trained and evaluated on identical, class-balanced data splits



PACE OCI's hyperspectral observations are modeled across space, spectrum, and time to estimate bloom risk, classify phytoplankton communities, and detect bloom events.

Limitations

- Our external validation attempt of testing the trained models against additional held-out in-situ samples was limited by **very small PACE-matched sample counts** (fewer than 15 usable matches), too few to properly evaluate whether these results hold up beyond the current dataset
- This means our results demonstrate within-dataset feasibility, not validated or generalizable performance
- This is a meaningful early signal, not a finished or operational model
- No previous approach to compare against.

Results

Best current bloom-risk detection:

87% accuracy, 0.81 macro-F1 for **small 3D Model.**

Best future bloom-risk estimation:

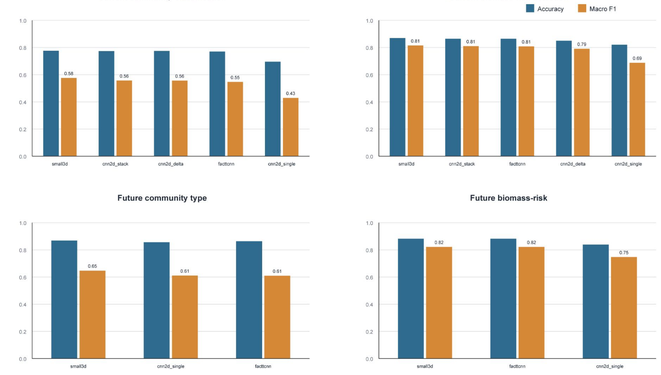
88% accuracy, 0.82 macro-F1 for **small 3D Model.**

Community classification: 78% accuracy, 0.58 macro-F1

These results show PACE reflectance carries a usable signal, especially for bloom-risk estimation, with the data available so far.

However, community classification's lower macro-F1 shows that class **imbalance & overlapping label categories** are affecting model performance.

Figure 2. Model comparison across current and future tasks



Macro-F1 gives equal weight to each class, making it more sensitive than overall accuracy to performance on rare or underrepresented categories.

Future Work

- Because in-situ data matched to PACE remains limited, we propose testing this same modeling framework on MODIS* as a way to evaluate whether the approach is feasible at a larger data scale
- If MODIS-based testing shows the framework generalizes well, that supports extending the approach back to PACE as more matched in-situ data becomes available over time
- In parallel, we are exploring novel ML approaches suited to small, diverse datasets, so the method isn't solely dependent on waiting for larger PACE matchups

**MODIS: a sensor with a longer historical record and denser matched in-situ coverage*